Ankit Kumar

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Education

•Birla Institute of Technology, Mesra *2026*
B.Tech in Electronics and Communication Engineering CGPA: 8.29

•Narbheram Hansraj English School *2022*
Indian School Certificate (ISC), New Delhi Percentage: 95

•Hill Top School *2020*
Indian Certificate of Secondary Education (ICSE), New Delhi Percentage: 94.2

Experience

•Samsung Research & Development, Noida *Jan 2026 - Jun 2026*
SDE Intern Noida

- Built a **Flask-ADB** diagnostic utility adding **illegal IMEI detection** at the customer service tier, removing engineering-escalation dependency and cutting diagnosis time to under 60 seconds; in staged rollout to **300+ Samsung service centers** nationwide.
- Rebuilt an internal file-transfer server (**Django**, **Nginx**) that hard-failed above 30 MB, implementing **chunked multipart uploads** with client-side segmentation, **async server-side reassembly**, and SQL-backed chunk-state tracking; removed the size cap, enabled **resumable uploads**, deployed across **5 teams serving 200+ users**.
- Designed a Carrier Aggregation extraction engine using **PyQt6 and Pandas** indexing 500+ config entries across 10+ Qualcomm spec files into a unified searchable interface, cutting engineer lookup time by **95%+** (5+ min to **under 10 sec**).

•Tata Motors *May 2025 - Jul 2025*
SDE Intern Jamshedpur

- Built a fault-isolated **ETL pipeline** aggregating 165+ Excel reports per 15-minute cycle from 11 edge sensor systems into a central store; engineered per-source **fault isolation** (file locks, network drops, machine-offline) ensuring zero-downtime ingestion.
- Built a real-time safety-compliance system (**YOLOv10**, Python) processing **5 live CCTV feeds** at **10 FPS** on standard laptop hardware, with automated **helmet-violation detection** and bounding-box audit logging, achieving **94%+ accuracy** on Bay 11 of the production line.

Projects

•QueryInsight : AI-Powered SQL Chatbot & Visualization Dashboard *May 2025*
Tools & technologies: Python, LangChain, Groq API, SQLAlchemy, MySQL, and Plotly/Dash.

- Engineered a multi-stage **NL-to-SQL** pipeline using **LangChain** and (**Groq API**) with dynamic schema injection via **SQLAlchemy** reflection, plus a three-stage validation layer (extraction, syntax check, schema cross-referencing) with read-only enforcement, and validation layer to eliminate invalid-column hallucinations before execution.
- Built an auto-selecting visualization layer in **Plotly/Dash** that infers chart type from query-result schema across **7 chart types**, routing aggregations to bar charts and time-series to line charts, with anomaly detection on returned datasets.

•Distributed Network Monitoring System *Jun 2026*
Tools & technologies :Java, Spring Boot, Apache Kafka, MongoDB, ShedLock, REST API, Docker

- Built a horizontally scalable, **fault-tolerant** distributed system with independent scheduler, worker, and ingestion roles (one **Spring Boot** app via profiles) over partitioned **Apache Kafka** topics, scaling out as **consumer groups**.
- Guaranteed single-active scheduling via a **MongoDB**-backed distributed lock (**ShedLock**) and **ordered, idempotent** processing through device-ID partition keying; containerized with **Docker**.

•Predictive Maintenance (RUL Prediction) *Feb 2025*
Tools & technologies: Python, TensorFlow, XGBoost, Pandas, Random Forest, Scikit-learn

- Built a **RUL prediction** system on **NASA C-MAPSS** (FD001), removing 7 redundant sensors via Pearson correlation with RUL clipping and 10-cycle rolling-mean augmentation, achieving **RMSE 32.48** (R2 0.71) with **SVR** and 4-fold cross-validation.
- Extended it to a **3-tier risk classifier** (No Risk, Moderate, Critical) using **SVC** and **Random Forest** with stratified splits, reaching **68%** validation accuracy across imbalanced classes.

Technical Skills

Languages: Java, Python, SQL
AI / ML: LangChain, Groq API, SQLAlchemy, Scikit-learn, XGBoost, SVR, Random Forest, YOLOv10, Plotly/Dash
Data Engineering: TL/ELT Pipelines, Apache Kafka, Data Modeling, SQL/NoSQL
MLOps: Docker, GitHub Actions CI/CD, AWS SageMaker (familiar), Model Monitoring
Databases: MySQL, MongoDB, BigQuery

Achievements

•Leetcode Knight (rating 1900+) *Jun 2026*

•Graduate Aptitude Test in Engineering (GATE) in DA Secured AIR 1388 *Feb 2025*

•Joint Secretary, Robolution Robotics Club *Feb 2024*